INCH - POUND

MIL-PRF-14516G
29 March 1999
SUPERSEDING
MIL-H-14516F
28 July 1993

PERFORMANCE SPECIFICATION

HEATER, SPACE - BARRACKS, TENT, NON-POWERED, LIQUID FUEL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance and acceptance requirements for a non-powered, liquid fuel burning, 60,000 BTU, radiant space heater for use in barracks, shelters, or tents.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 4520

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STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-810 - Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of the above standard are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to a first article inspection in accordance with 4.1.1.

3.2 Conformance. A sample shall be subjected to conformance inspection in accordance with 4.1.2.

3.3 **Operating requirements**. The heaters shall satisfy the following user-oriented requirements.

3.3.1 Operation. The heater shall operate with all specified fuels and under all specified conditions herein with no electrical power.

3.3.2 Ignition. The heaters shall not require any electrical power for ignition. Ignition shall be accomplished safely in ten minutes or less.

3.3.3 Monitoring. The heaters shall incorporate a means for viewing the entire combustion flame during operation.

3.3.4 Liquid fuel. The heaters shall be safely operated with all of the fuels listed in Table I.

TABLE I. Compatible Fuels

Military Symbol	Ambient Temperature Range (°F)
DF-2	above 25 °F
DF1	above 10 °F
DFA	above –60 °F
JP8	above –60 °F

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3.3.5 Fuel supply. The fuel supply shall be from the standard high density polyethylene 5-gallon military fuel can (see 3.6.2), mounted on the heater and delivered to the fuel control valve by means of a siphon arrangement. The can mounting area shall have a means to protect the fuel can from heat damage and prevent the fuel from reaching its flash point temperature (JP8-100°F).

3.3.6 Fuel control. The heaters shall have a fuel control that maintains a constant flow rate to the burner from the 5-gallon can. The fuel control shall have a metering knob to control flow rate incrementally with a minimum of 5 settings. The fuel control shall provide sufficient fuel flow to achieve the full range of required heat outputs corresponding to table II utilizing no less than 70% of the full range of settings on the valve. The fuel control shall be permanently labeled to identify settings. The fuel control shall control fuel flow to the burner to eliminate any possibility of an unsafe over fire condition when operated at the proper settings for the prevailing ambient temperatures and fuel types. The fuel control shall regulate the firing rate in conjunction with the heater, such that, for over 70% of the full range of settings, smoke levels shall not exceed a # 4 on the Bacharach smoke scale.

3.3.7 On/off switch. The heater shall have an on/off switch in an easily accessible location which immediately cuts off fuel to the burner when turned off.

3.3.8 Fuel filtering. A means to eliminate foreign particles from obstructing fuel flow shall be included. It shall be serviceable within 15 minutes with common hand tools with out the need for supplemental sealers to prevent leaks.

3.3.9 Combustion efficiency. The heater shall have a combustion efficiency greater than 60% when operated 50,000 BTU/hr or higher.

3.3.10 Smoke. The smoke reading of the heater shall not be greater than a No. 4 on the Bacharach smoke scale when the heater is operated between 30,000 BTU/Hr and 60,000 BTU/Hr.

3.3.11 Heat output. The heater shall provide the BTU outputs listed in table II.

TABLE II. Heat Output Requirements

Ambient (E F)	Fuel	Output (low)	Output (high)
40	DF2	30,000 BTU/Hr	50,000 BTU/Hr
-25	JP8	40,000 BTU/Hr	60,000 BTU/Hr

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3.3.12 Exhaust pipe. The heater shall be supplied with either a type I or type II exhaust pipe assembly to vent all products of combustion outside the shelter, as specified in the contract or purchase order (see 6.2). The heater shall meet all operating requirements with either exhaust pipe configuration.

3.3.12.1 Type I exhaust pipe. The type I exhaust pipe shall be nestable for storage within the heater envelope. When assembled on the heater, the outside diameter of the exhaust pipe shall be no greater than 5 inches at a height of 105 inches from the ground in order to easily pass through the exhaust pipe opening on the General Purpose (GP) Medium tent. The exhaust pipe shall be within 145-160 inches tall when assembled onto the heater. A means to prevent rain from entering the exhaust pipe and a means to stabilize the exhaust pipe shall be included.

3.3.12.2 Type II exhaust pipe. The type II exhaust pipe assembly shall consist of eight 6" diameter sections 24" long each. An exhaust pipe cap shall also be included. If necessary a means to maintain proper draft for operation shall be included. When specified (see 3.3.12.3 and 6.2), a smoke jack assembly to rout the exhaust outside of a wooden roof shall be supplied

3.3.12.3 Smoke jack assembly. When specified (see 6.2 and 6.8), a means to safely route the exhaust pipe through a ceiling and wooden roof shall be provided. When properly assembled, there shall be no rain leakage through the roof jack assembly. The assembly shall be adjustable in pitch to accommodate varying roof angles. An insect screen shall be included to prevent intrusion into the shelter when stove pipe is disassembled. The smoke jack shall be capable of installation using common hand tools. The smoke jack assembly shall be securely attached to the roof when installed. The smoke jack shall provide sufficient insulation from the hot exhaust to prevent heat damage to the wooden ceiling and roof.

3.3.13 Tilt. The heater shall operate at a two degrees tilt without causing a safety hazard or performance degradation.

3.3.14 Safety. The heater and its components shall perform all modes of operation in a safe manner. All equipment shall include fail safe features for safety of personnel during installation, operation, maintenance, and repair activities. The heater configuration shall eliminate the possibility of improper assembly of components (i.e. burner assembly, fuel control) which may lead to a hazardous condition.

3.3.14.1 Fuel flow. In the event of flame loss, fuel shall automatically stop entering the burner area.

3.3.14.2 Burner. There shall be no pool of raw fuel in the burner during operation or in the event of flame failure.

3.3.14.3 Exhaust. Products of combustion shall remain separated from the breathable heated air.

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3.3.14.4 Sharp edges. Non-functional sharp edges and projections shall be eliminated.

3.3.14.5 Noise. Noise levels, as measured from 3 feet from the heater, shall not exceed 60 dB.

3.3.14.6 Bottom surface temperature. The bottom surface of the heater in contact with the floor shall remain at 175E F or less during all modes of operation.

3.4 **Support or ownership requirements**. Each heater shall meet the following logistics requirements.

3.4.1 Mean Time Between Essential Function Failure (MTBEFF)/Maintenance Ratio(MR). The heaters shall meet a quantitative reliability requirement MTBEFF of 615 hours . The heater maintainability requirement is a total unit, direct support (DS), and, general support (GS) MR which shall not to exceed 0.0016 maintenance man-hour per operating hour, not including preventive maintenance checks and services (PMCS).

3.4.2 Preventive maintenance checks and services (PMCS). . PMCS shall not exceed 10 minutes time to accomplish. The heater fuel lines and burner shall be designed in such a manner to allow for operator inspection. The minimum PMCS items to be checked shall be secure connections of lines, fuel leaks, exhaust obstructions, burner, and general condition to permit operation. There shall be no new system specific test, measurement, and diagnostic equipment, or special tools to repair or maintain the heater.

3.4.3 Transportability. The heaters shall be transportable in the High Mobility Multipurpose Wheeled Vehicle (HMMWV), heavy-HMMWV, the Commercial Utility Cargo Vehicle (CUCV), and standard Army cargo trucks (2 **2** to 5 ton) and all standard Army cargo trailers having a **3** to 1 **2** ton capacity.

3.4.3.1 Loose cargo. The heaters shall be of sufficient durability to be transported as loose cargo in any of the required transport vehicles.

3.4.3.2 Drop . The heaters shall be of sufficient durability to survive an accidental drop from any of the required transport vehicles.

3.4.4 Drag capability. The heater shall be capable of being dragged across a flat surface by one soldier.

3.4.5 Labels/tags. The heaters shall have an instruction plate permanently attached in an immediately obvious location which provides warnings, start up and shut down procedures, operating procedures, and safety cautions. The fuel control and heater lettering shall be permanent for the life of the item. Controls shall not obscure labels. Labels shall be located in a

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consistent manner throughout the system. Labels shall be printed in capitals. Letter size and width shall be sufficient to be easily read from a distance of six feet in daylight or with the aid of a flashlight.

3.4.6 Weight. The weight of the heater without accessories shall be less than 80 lbs

3.4.7 Volume. The outside dimensions of the heater shall not exceed 8.6 cubic feet

3.4.8 Lifting. The heaters shall have handles and be transportable by two people wearing cold weather gear.

3.4.9 Personnel clothing. The heaters shall be easily operated and set up by personnel attired in clothing appropriate for all intended user environments. The heater shall be capable of being assembled for operation in 20 minutes or less, and disassembled and repacked for storage by two people in 15 minutes or less in any climatic category specified herein.

3.5 **Environmental requirements**. The heater shall operate as specified under the following environmental conditions.

3.5.1 Storage. The heater and accessories shall remain operable and show no signs of impairment due to storage at 155E F for 4 hours daily.

3.5.2 Basic climate. The heater and accessories shall be operable in temperatures of +60E F to 0E F. Heat outputs shall be in accordance with Table II.

3.5.3 Basic cold. The heater and accessories shall be in temperatures of -5 F to -25E F. Heat outputs shall be in accordance with Table II.

3.5.4 High altitude. The heaters shall operate safely at a minimum of 85% rated capacity when operated at altitudes of 4,000 feet above sea level.

3.6 **Interface requirements**. The heater shall meet the following interface and interoperability requirements.

3.6.1 Thermo-electric fan interface. The heater shall accommodate placement of the thermo-electric fan unit (see figure 1), so that it is positioned to direct air downward. An indented area shall be provided for placement of the base of the fan (see figure 2). The indented area shall provide sufficient heat transfer to the fan base to achieve a temperature of 500EF - 600EF at the fan module surface. If necessary, a spacer shall be permanently located on the indented area to position the fan base above the heater surface to prevent over heat damage, while maintaining enough heat to achieve the required module temperature.

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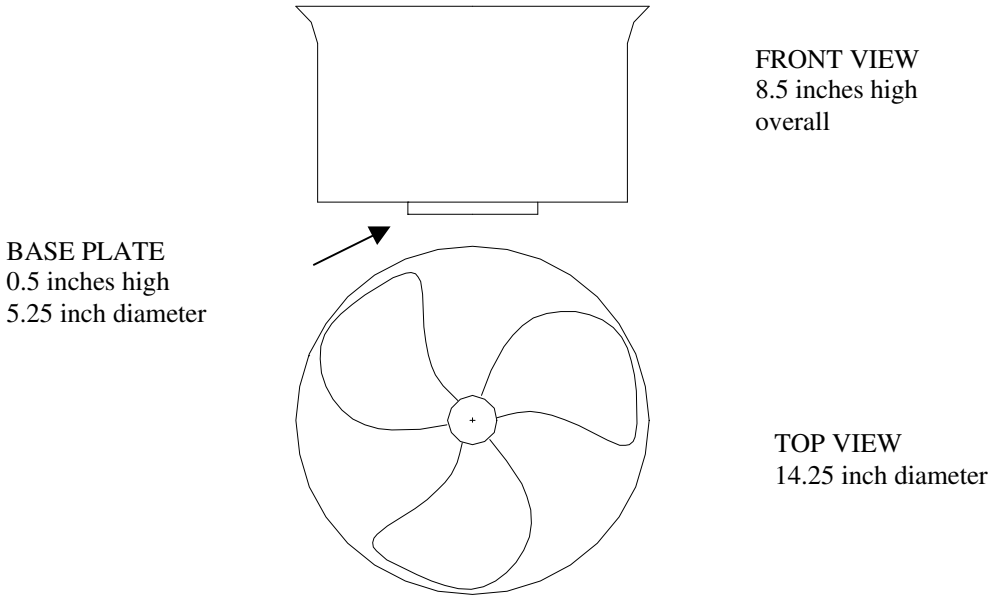


FIGURE 1. Thermo-Electric Fan

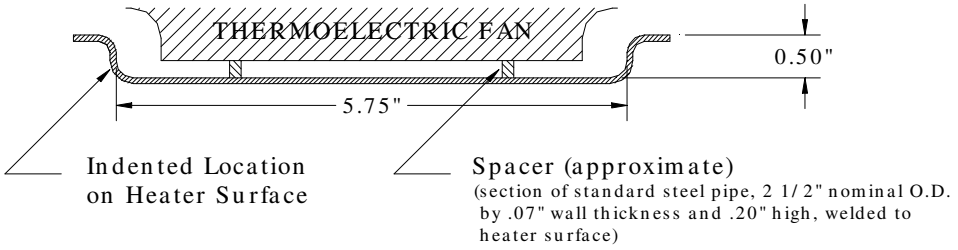


FIGURE 2. TEF interface location and typical spacer.

3.6.2 Fuel interface. During test of 4.5.2 the heater shall operate on DF2 conforming to A-A-52557. During test of 4.5.3, the heater shall operate on JP8 conforming to MIL-DTL-83133. During all operational tests herein, the fuel can used for the fuel supply shall conform to MIL-C-53109.

3.6.3 Tent interface. The heaters shall be compatible for operation in the General Purpose Medium tent, MIL-T-1712.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.1.1).
b. Conformance inspection (see 4.1.2).

4.1.1 First article inspection. When a first article inspection is required (see 3.1 and 6.2), the heater shall undergo the tests and examinations outlined in table III. Sampling for first article inspection shall be as specified in the contract or purchase order.

4.1.2 Conformance inspection. Conformance inspections shall be conducted according to the tests specified in table III. Sampling for conformance inspection shall be as specified in the contract or purchase order (see 3.3 and 6.2).

4.1.3 End item examination. Each heater shall be examined for the requirements specified in Table III. Any redesign or modification following failure to meet the specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements.

4.2 Verification methods. Verification methods can include visual examination, measurement, testing, simulation, modeling, engineering evaluation, component properties analysis, certification, and similarity to previously approved or previously qualified designs.

4.2.1 Verification alternatives. The contractor may propose alternative test methods, techniques, or equipment, including application of statistical process control, tool control, or cost-effective sampling procedures to verify performance.

TABLE III. Requirement and Verification Outline

Requirement	Requirement Paragraph	Verification Paragraph	First article inspection	Conformance inspection	End item Examination
Operating Requirements	3.3	4.3			
Operation	3.3.1	4.3.1	X	X	
Ignition	3.3.2	4.3.2	X	X	
Monitoring	3.3.3	4.3.3	X	X	
Liquid fuel	3.3.4	4.3.4	X	X	
Fuel supply	3.3.5	4.3.5	X		
Fuel control	3.3.6	4.3.6	X	X	
On/Off switch	3.3.7	4.3.7	X	X	

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Fuel filtering	3.3.8	4.3.8	X	X	
Combustion efficiency	3.3.9	4.3.9	X		
Smoke	3.3.10	4.3.10	X		
Heat output	3.3.11	4.3.11	X		
Exhaust pipe	3.3.12	4.3.12	X		
Type I exhaust pipe	3.3.12.1	4.3.12.1	X		
Type II exhaust pipe	3.3.12.2	4.3.12.2	X	X	
Smoke jack assembly	3.3.12.3	4.3.12.3	X	X	
Tilt	3.3.13	4.3.13	X		
Safety	3.3.14	4.3.14	X		
Fuel flow	3.3.14.1	4.3.14.1	X		
Burner	3.3.14.2	4.3.14.2	X		
Exhaust	3.3.14.3	4.3.14.3	X		
Sharp Edges	3.3.14.4	4.3.14.4	X	X	X
Noise	3.3.14.5	4.3.14.5	X		
Bottom surface temperature	3.3.14.6	4.3.14.6	X		
Support or ownership requirements	3.4	4.4			
MTBEFF/MR	3.4.1	4.4.1	X		
PMCS	3.4.2	4.4.2	X		
Transportability	3.4.3	4.4.3	X		
Loose cargo	3.4.3.1	4.4.3.1	X		
Drop	3.4.3.2	4.4.3.2	X		
Drag capability	3.4.4	4.4.4	X		
Labels/tags	3.4.5	4.4.5	X	X	
Weight	3.4.6	4.4.6	X		
Volume	3.4.7	4.4.7	X		
Lifting	3.4.8	4.4.8	X		
Personnel clothing	3.4.9	4.4.9	X		
Environmental requirements	3.5	4.5			
Storage	3.5.1	4.5.1	X		
Basic climate	3.5.2	4.5.2	X		
Basic cold	3.5.3	4.5.3	X		
High altitude	3.5.4	4.5.4	X		
Interface requirements	3.6	4.6			
Thermo-electric fan	3.6.1	4.6.1	X		

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interface					
Fuel interface	3.6.2	4.6.2	X		
Tent interface	3.6.3	4.6.3	X		

4.3. **Operating requirements verification.** All operating requirements shall be verified as follows.

4.3.1 Operation inspection. Verify proper operation during all performance and reliability testing specified herein. Any use of electrical power for operation shall constitute a failure.

4.3.2 Ignition inspection. Observe heater on start. Failure of heater to operate using a match as an ignition source shall constitute a failure. Failure of the heater to be operating within specified smoke and efficiency readings within 15 minutes after lighting under any environmental condition specified herein, shall constitute a failure.

4.3.3 Monitoring inspection. Observe the heater during operation. Inability to view entire flame through a sight glass or similar means shall constitute a failure.

4.3.4 Liquid fuel verification. Verify fuel compatibility during environmental climate testing of 4.5.2, and 4.5.3. Verify fuels do not degrade any heater components such as fuel lines and gaskets during reliability testing.

4.3.5 Fuel supply inspection. During environmental and reliability testing determine adequacy of siphon arrangement for safe usage, visually inspect fuel can for signs of heat damage and measure fuel temperature at all times to verify safe temperature below flash point.

4.3.6 Fuel control test. Determine adequacy of the fuel control during environmental climate testing of 4.5.2, and 4.5.3. Operate the heaters over the full range of settings. Measure heat output as outline in Appendix A for each setting. Failure of the flow rate and heat output to incrementally increase shall constitute a test failure. Measure the smoke level per Appendix A over the range of settings during the climatic tests of 4.5.2, and 4.5.3. Any smoke readings greater than #4 for over 30% of the range of settings shall constitute a failure. Presence of an unsafe over fire condition during testing as evidenced by black smoke exiting the exhaust stack, pulsing, sputtering or backfiring of the heater or excessive sooting shall constitute a failure. During test of 4.3.2, measure flow rate to the heater at setting 4 after 1 hour of operation and record. After one more hour of operation at a constant ambient temperature with the tent doors closed and at a fuel control setting 4, measure the flow rate to the heater. An increase greater than 10% over the previous reading shall constitute a failure.

4.3.7 On/off switch test. Connect a buret in-line with a gravity feed fuel supply line to the fuel control. After the heater has been operating for a minimum of 10 minutes, turn the on/off

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switch to the off position and observe flow rate in the buret. Failure of fuel flow to stop immediately upon activation of the off switch shall constitute a failure.

4.3.8 Fuel filtering inspection. Observe condition of filtering device after reliability testing. Any heater shut downs related to particles bypassing the fuel filtering device or the device rusting and preventing fuel flow shall constitute a failure. Failure to remove and replace the filtering device for cleaning in less than 15 minutes using only common hand tools shall constitute a failure.

4.3.9 Combustion efficiency test. During environmental testing of 4.5.2, and 4.5.3, after durability testing, every 24 hours during reliability testing, and as required, measure efficiency per procedure outlined in Appendix A. Any readings lower than 60% while the heater is operating at 50,000 BTU/Hr or higher shall constitute a test failure.

4.3.10 Smoke test. During environmental testing of 4.8.2 and 4.8.3, after durability testing, and every 24 hours during reliability testing, measure smoke readings per procedure outlined in Appendix A. Any readings greater than a #4 during any required firing rate shall constitute a failure.

4.3.11 Heat output test. See 4.5.2, 4.5.3

4.3.12 Exhaust pipe inspection. Inspect for presence of type I or type II exhaust pipe assembly.

4.3.12.1 Type I exhaust pipe inspection. Inspect heater during operation and verify that the exhaust pipe will pass through the opening in the GP medium tent. Verify the presence of an exhaust pipe cap with guy lines or other means to keep exhaust pipe from blowing off in windy conditions.

4.3.12.2 Type II exhaust pipe inspection. Inspect for the complete exhaust pipe assembly per 3.3.12.1, including smoke jack assembly when specified (see 6.2).

4.3.12.3 Smoke jack assembly inspection. Inspect for complete smoke jack assembly per 3.3.12.3.

4.3.13 Tilt test. Operate the heater. After the heater has stabilized tilt the heater 2 degrees and operate for a minimum of 15 minutes. Failure of the heater to continue normal safe operation shall constitute a failure. Perform this test a total of four times, tilting the heater in both directions on the two bottom orthogonal axes.

4.3.14 Safety inspection. Observe test records and performance of all test and inspections within this performance specification. Inspect heater system documentation for adequacy of

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safety information. Presence of any hazards, injury to personnel or damage to equipment or property resulting from testing shall constitute a failure.

4.3.14.1 Fuel flow inspection. Verify fuel flow stops entering the combustion area during a flame loss condition.

4.3.14.2 Burner inspection. Verify no raw fuel accumulates in the combustion area of the heater after shut off or during operation.

4.3.14.3 Exhaust test. Test for, at a minimum, carbon dioxide, carbon monoxide, ammonia, nitrogen dioxide and formaldehyde while the heater is set up and operated in a GP Medium tent using the Type I exhaust pipe in an ambient temperature of 40E F. The heaters shall operate for a minimum of 8 hours and all doors shall be closed during testing with no one entering the tent during the test. Any readings above the TLV-TWA as established by the ACGIH shall constitute a failure. The test shall be conducted using JP-8 fuel.

4.3.14.4 Sharp edges. Visually inspect the heater. Presence of any non-functional sharp edges shall constitute a failure.

4.3.14.5 Noise. Operate the heater at the high fire rate. Measure sound levels at a 3-foot distance around the outside of the heater on four sides and the top. Any readings in excess of 60 dB shall constitute a failure.

4.3.14.6 Bottom surface temperature inspection . Inspect temperature of base of heater in contact with the floor during operation. Any readings in excess of 175E F shall constitute a failure.

4.4 **Support or ownership requirements**. All support or ownership requirements shall be verified as follows.

4.4.1 Mean Time Between Essential Function Failure (MTBEFF)/Maintenance Ratio (MR) test. Four heaters shall each be placed on a reliability test for a minimum of 461 hours each for a total of 1844 hours collective. Testing shall be conducted at 68E F and the heaters shall be at a setting to maintain 50,000 BTU/Hr. There shall be no more than one chargeable essential function failure (cumulative). Fuel type used for test shall be divided equally among DF-2 and JP-8. Perform burner smoke spot, during the start and each 100 hour interval (" 10 hours) and once during the last start cycle at the conclusion of this test for each unit operated. A maintenance ratio greater than .0016 maintenance man-hour per operating hour for any maintenance other than PMCS shall constitute a test failure. . An essential function failure (EFF) shall be scored when an incident results in the loss or significant degradation of an essential function that can not be quickly (within 15 minutes) corrected by the crew using only on board tools, spares and repair parts. An incident that causes catastrophic or critical hazards shall also be scored as an EFF.

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4.4.2 Preventive maintenance checks and services (PMCS) inspection. Configure the heater for operation. Perform a walk around inspection of the heater, verify a secure fuel control connection, the fuel supply for proper connection, fuel leaks, and general operating condition. A thorough examination shall not exceed 10 minutes.

4.4.3 Transportability tests. Verify the heater is no higher than 35 inches tall.

4.4.3.1 Loose cargo test. The heater shall be subject to loose cargo testing IAW the requirements and conditions set forth in MIL-STD-810, Method 514.4, Category 3, Procedure III.

(1) Visually check all contents and functionally check the heater.

(2) Place the heater, outside of it's original packaging box, on a tester using fences and a plywood bed configured IAW MIL-STD-810. The package tester speed shall be 300 " 2 RPM. Total test time of 45 minutes shall be divided into the four major loading configurations by stopping the test and rotating the heater 90 degrees and continuing the test as needed.

(3) Upon completion of the 45 minutes of testing, perform a visual and a functional check of the heater and components.

Any component damage, which prevents the heater from operating after loose cargo testing, or any damage on the heater or accessories which may lead to unsafe operation shall constitute a test failure.

4.4.3.2 Drop testing. The heater shall be subject to drop testing as follows:

(1) Visually check all contents and functionally check the heater.

(2) Suspend the heater, outside of it's original packaging box, by a drop hook such that the desired edge, corner or face is 24 inches from the impact surface. The drop hook shall be triggered allowing the heater to free fall to the drop surface.

(3) The drop surface shall be a packed dirt surface.

(4) Perform a total of four drops on the bottom edge, at four different impact surfaces located 90 degrees apart.

(5) Upon completion of the drop testing, perform a visual and operational check.

Any component damage which prevents the heater from operating after drop testing or any damage on the heater or accessories which may lead to unsafe operation shall constitute a test failure.

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4.4.4 Drag capability inspection. Verify that one person can easily drag the stove when hot.

4.4.5 Labels/tags inspection. Any signs of labels wearing out from normal usage during reliability testing shall constitute a test failure. Omission of any of the required labels shall constitute a failure.

4.4.6 Weight inspection. Weigh the heater without accessories. Any weight readings exceeding the requirements of 3.4.6 shall constitute a failure.

4.4.7 Volume inspection. Measure the outside envelope dimensions of the heater. Any measurements greater than the requirements listed in 3.4.7 shall constitute a failure.

4.4.8 Lifting inspection. Verify the heater can be lifted and moved by two 80th percentile soldiers a distance of 100 feet and lifted to a height of 36 inches by the handles.

4.4.9 Personnel clothing inspection. Verify heater can be configured for operation, started ,and controls can be manipulated while wearing cold weather gloves and cold weather clothing during test of 4.5.3. Set up shall be conducted by no more than two people. Average set up times greater than 20 minutes for each heater shall constitute a failure. Repack each heater for storage. Disassembly times greater than 15 minutes for each heater shall constitute a failure.

4.5 **Environmental requirements**. All environmental requirements shall be verified as follows.

4.5.1 Storage test. Subject one heater to storage testing. Bring chamber air temperature up to 155E F (" 3EF) and 75% relative humidity (" 5%), once these conditions are attained, hold for a 4 hour storage period. Remove the heater and perform functionally check. Failure of the heater to operate due to damage incurred during storage testing shall constitute a test failure.

4.5.2 Basic climate test. Stabilize climatic chamber ambient air temperature at 40E F (" 2EF) to operate heater. Cold soak heater, accessories and fuel. Set up and operate heater in the GP small tent or similarly sized shelter with an exhaust stack opening. Use fuels specified (see Table II). Perform indicated efficiency, smoke test and heat output test per Appendix A. Failure of the heater to operate over the minimum range of 30,000 BTU/Hr - 60,000 BTU/Hr within acceptable smoke and efficiency levels shall constitute a failure.

4.5.3 Basic cold test. Stabilize climatic chamber air temperature at -25E F ("2EF) to operate heater. Cold soak heater, accessories and fuel. Unpack heater and set up for operation as in 4.5.2. Use fuels specified (see Table II). Perform indicated efficiency, smoke test and heat output test per Appendix A. Failure of the heater to operate over the minimum range of 30,000 BTU/Hr - 60,000 BTU/Hr within acceptable smoke and efficiency levels shall constitute a failure.

4.5.4 High altitude test. Provide a field site, which is 4,000 feet Above Sea Level (ASL) minimum, or simulate conditions at 4,000 feet ASL for operational checkouts. Operate heater for a

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minimum of one hour. Ambient temperature shall be 50E F or below. Perform smoke spot per Appendix A within 15 minutes after starting and again before shut down. Perform indicated efficiency test per Appendix A and measure fuel flow. Heat output less than 85% rated capacity, unacceptable smoke readings, heater pulsing, or sputtering, shall constitute a failure.

4.6 **Interface requirements inspection**. All interface and interoperability requirements shall be verified as follows.

4.6.1 Thermo-electric fan interface. A thermo-electric fan will be supplied as Government Furnished Property (GFP) to conduct this test. Place the thermo-electric fan on the heater designated location during operation. Failure of the TEF to fit evenly on the heater designated surface such that heated air is directed downward shall constitute a failure. In an ambient temperature between 60E F and 80 E F, operate the heater at 60,000 BTU/hr output. Place the TEF on the designated location. After 30 minutes of operation, with the fan still operating on the heater, measure the module temperature using a handheld digital thermometer capable of reading K-type thermo-couples from the miniature panel jack mounted on the TEF. Failure to maintain a temperature between 500E F to 600E F shall constitute a failure.

4.6.2 Fuel interface inspection. During tests of 4.5.2 and 4.5.3 verify that fuels used are as specified in 3.6.2.

4.6.3 Tent interface inspection. During tests of 4.5.2 and 4.5.3, verify that tent is as specified in 3.6.3. Verify exhaust system is compatible with stove pipe opening in tent.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2). When actual packaging of material is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Departments System Command. Packaging data retrieval is available from the managing Military Departments or Defense Agency's automated packaging files, CD-ROM products, or by contracting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory)

6.1 Intended use. The heaters are intended for use primarily for heating prefabricated and temporary-type buildings and large tents. This item is military unique due to the requirements of

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multi-fuel operation, operation without electrical power, interfacing with existing military shelters, mobility, and ruggedness.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).
c. When a first article is required (see 3.1.).
d. Type of exhaust assembly required (see 3.4.12 and 4.4.12)
e. When a smoke jack assembly is required (see 3.4.12.2 and 4.4.12.2)
f. Sampling plan for first article and conformance test (see 4.2 and 4.3).
g. Packaging requirements (see 5.1).
h. Operators manual.
i. Replacement parts certification of interchangeability.

6.3 Subject term (key word) listing.

Diesel fired space heater
JP-8 fired space heater
Pre-fabricated shelter heating
Tent heating

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals should be acquired under separate contract line item in the contract.

6.5 Verification alternatives. Contracting documents should provide guidance to offerors regarding the submission of alternatives to specified verification methods (see 4.3.3).

6.6 First article. When requiring a first article, contracting documents should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, a first production item, and the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article results, and disposition of first articles. Pre-solicitation documents should provide Government waiver rights for first article inspection to bidders offering a previously acquired or tested product. Bidders offering such products who wish to rely on previous testing must furnish evidence with the bid that prior government approval is appropriate for the pending contract.

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6.7 Recycled, recovered or environmentally preferable materials. Recycled, recovered or environmentally preferable materials should be used to the maximum extent possible, provided the materials meet or exceed the operational, environmental and maintenance requirements specified herein and promote economically advantageous life cycle costs.

6.8 Smoke jack assembly. The smoke jack assembly (NSN 4520-00-999-0676), drawing number 13208E6218, has been found to satisfy the requirements of 3.3.12.3

Custodians:
Army - GL
Navy - YD1
Air Force - 99

Review activities
Army - CR4
Navy - MC
Air Force - 84
DLA-IS

Preparing activity:
Army - GL

(Project 4520-0428)

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APPENDIX A

Performance Test Procedures

A.1 SCOPE

A.1.1 Scope. This appendix details the performance testing procedures for measuring indicated combustion efficiency, heat output, and burner smoke spot; and is a mandatory part of this specification.

A.2 APPLICABLE DOCUMENTS. None.

A.3 PROCEDURE

A.3.1 Heater test set up. For all performance testing, the following minimum conditions apply:

a. The heater shall be set up in the normal operating configuration with the correct length of exhaust pipe and on a level surface inside the proper tent.

b. A sampling hole (approximately **3** " dia) shall be drilled in the exhaust pipe below the draft regulator (if used). The draft shall be natural with out the use of any mechanical draft inducer.

c. Connect a 100ml buret in-line with a gravity feed fuel supply line to measure fuel flow instead of the required siphon arrangement.. An Alltech Associates, Inc. 100cc flowmeter stock # 4045 or equal is sufficient for this test.

A.3.2 Combustion efficiency test. The heaters shall be operated for a minimum of 15 minutes at the required setting before taking readings. The following readings shall be taken consecutively within five minutes:

a. Measure flue gas temperature at the sampling hole, a thermometer capable of measuring up to 1300 F shall be used.

b. Measure air temperature immediately around the combustion air intake area of the heater.

c. Measure the CO2 % from the sampling hole using a Bacharach CO2 indicator, Part No. 11-7032.

d. Use the following formula for determining the combustion efficiency:

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$$E = 1 - \frac{3171 + 2.69T_s + \frac{216.2 + .77}{CO2}(T_s - T_I) - 5.38\,T_i}{HHV}$$

E = Combustion efficiency (%/100)
TS = Stack temperature (Degrees C)
TI = Combustion air inlet temperature (Degrees C)
CO2 = Carbon dioxide (%)
HHV = High heat value (joules/gram) (from fuel certification)

A.3.3 Heat output.

Heat output (BTU/Hr) = F X P X (HHV/1055) X E X 60

F = Fuel flow rate (ml/min)
P = Fuel density (gm/ml) (DF-2 = .852 gm/ml)
HHV = High heat value (DF-2 = 45166 joule/gram)
E = Combustion efficiency (from A.2.2)

A.3.4 Smoke spot test. At the sampling hole, measure the smoke spot using a Bacharach Fyrite test kit, P/N 10-5000, smoke tester and compare to the Bacharach smoke spot scale. The heater shall be operating for a minimum of 15 minutes before taking the smoke sample.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-14516G	**2. DOCUMENTDATE (YYYYMMDD)** 990329

DOCUMENT TITLE ELECTRODES, WELDING, COVERING: AUSTENITIC STEEL (19-9 MODIFIED) FOR ARMOR APPLICATIONS

4. NATURE OF CHANGE*(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN *(if applicable)*	**7.DATE SUBMITTED** (YYYYMMDD)

8. PREPARING ACTIVITY

a. NAME U.S ARMY SOLDIER AND BIOLOGICAL CHEMICAL COMMAND	b. TELEPHONE *Include Area Code)* (1) Commercial (508) 233-5592 (2) DSN 256-5592
c. ADDRESS *(Include Zip Code)* ATTN: AMSSB-PM-RSS-F(N) Kansas Street Natick, MA 0176-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE STANDARDIZATION PROGRAM OFFICE (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, Virginia 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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